Idea 19+ — The Assurance Gradient: Certification-Gap Analysis Across Voting Machines, Electronic Poll Books, and Voter Registration Systems

Status: SELECTED, pre-design. No data collected, no analysis run. This document is a research plan, not a paper.

Policy context (verified 2026-08-30) — why this study is time-critical

The federal election-assurance infrastructure this study analyzes is currently inoperative at the leadership level:

- **EAC has zero commissioners.** Donald Palmer resigned 2026-04-30; the remaining commissioners (Thomas Hicks, Benjamin Hovland) were dismissed 2026-07-09; as of July 2026 the commission has no members and no quorum (quorum = 3 of 4). (Ballotpedia; Institute for Responsive Government, 2026-07-10; CDT, 2026-07-10; Bipartisan Policy Center, 2026-05-07)
- **CISA election security has been gutted.** ~130 election-security staff cut in Feb 2025 (DOGE); EI-ISAC funding terminated; the FY2027 budget proposal eliminates CISA's election-security program entirely. (Brennan Center, 2025-04-14; Nextgov, 2026-04-28; MS NOW, 2026-07-22)

Consequences for the study design: (a) VVSG adoption, new certifications, and RfI resolution are frozen or uncertain at the commission level (staff-level testing historically continues through vacancies, but no commission exists to act); (b) the ESTEP/EPB program's future is uncertain mid-flight; (c) states are actively discussing "Plan B" assurance arrangements (CDT, above). The public documentary record of the *existing* regime — VVSG text, VSTL reports, RfIs, ESTEP pilot materials — is complete, stable, and lawfully obtainable now, regardless of the institutions' future.

Policy + technical framing (two contributions, one study)

- **Technical contribution:** the requirements→test→evidence→failure traceability graph, coding schema, reliability statistics, and measured coverage gaps, exactly as specified below.
- **Policy contribution:** the same artifact functions as an evidence base for the post-EAC-quorum landscape — it tells states, courts, and any successor body *what the certification regime demonstrably did and did not cover*, which failure classes were never testable under it, and what a state-run or reconstituted federal program would need to add. In a period with no commissioners and no CISA election program, an independently reproducible, public-evidence assurance audit is not just a measurement study; it is the only form of federal-level certification scrutiny currently

operating. Frame the paper accordingly: methods + measurement in the body, implications-for-assurance-governance in the discussion. Candidate policy-facing outlets for a companion piece: IEEE Security & Privacy magazine, CRA CCC white paper, Election Law Journal.

One-paragraph summary

US election technology sits at three radically different assurance levels: voting machines are federally certified (VVSG 1.0/2.0 via EAC/VSTLs), electronic poll books (EPBs) have only a voluntary EAC program (ESTEP, launched May 2024, draft requirements VEPBCR 1.0), and statewide voter registration (VR) systems have no certification or requirements framework at all. This study builds a machine-checkable requirements→test→evidence→failure traceability graph across all three tiers from public documents only, and measures which documented real-world failure classes each tier's assurance process could and could not have detected.

Research question

What measurable fraction of assurance requirements across the three election-technology tiers has public test evidence that would detect the failure classes documented in court-filed expert reports, CISA advisories, and verified public incidents?

Hypothesis (falsifiable)

A large class of demonstrated failures — including the scalable code-execution findings in the Halderman/Springall ImageCast X report and documented EPB/VR weaknesses — maps to no failing requirement and no public test evidence in the tier that governs the component; and the least-governed tier (VR) is the one the other two tiers transitively trust.

Core contribution

1. First peer-reviewed security/certification-gap analysis of electronic poll books (timed with the ESTEP/VEPBCR program).
2. First cross-component (machine + EPB + VR) assurance traceability methodology with published coding schema and inter-rater reliability.
3. A measured, two-directional result: either certification gaps are structural and quantified, or certification is shown stronger than critics claim. Both outcomes are publishable.

Mechanism

- Corpus: public EAC/VSTL test reports, VVSG 1.0/2.0 requirement text, Requests for Interpretation, state test reports (e.g., CA SOS

Robis EPB review 2018; NJ/NC/IN protocols), VEPBCR 1.0 draft + ESTEP pilot report (2023, 7 EPBs), CISA ICSA-22-154-01, the unsealed Halderman/Springall report (2023), DEF CON Voting Village reports (as gray literature, labeled as such), NASEM 2018, documented VR incidents (Sweeney et al. 2017 classes; 2024 Georgia cancellation-portal disclosure).

- Encode failure classes as testable predicates; link requirement→test-procedure→public-evidence→failure-class in a graph; double-code a sample with documented adjudication; report Cohen's .
- Gap metric: fraction of failure classes with no detecting requirement/test per tier; VVSG 1.0→2.0 delta; evidence-opacity metric where documents are too redacted to classify (opacity itself is a finding).

Closest prior work (verified)

- California Top-to-Bottom Review (2007); Ohio EVEREST (2007) — one-off reviews, no repeatable methodology.
- Halderman & Springall, ImageCast X analysis (2021, unsealed 2023, Curling v. Raffensperger) + CISA ICSA-22-154-01 — failure-class ground truth; EAC stated no certification requirement was violated.
- EAC ESTEP EPB program + VEPBCR 1.0 draft (verified on eac.gov) — the EPB requirements target.
- Cable, Fábrega, Park & Specter, "A Systematization of Voter Registration Security," J. Cybersecurity 9(1):tyad008 (2023) — VR SoK; MUST cite.
- Sweeney, Yoo & Zang, voter identity theft study, J. Technology Science (2017) — VR portal weaknesses; MUST cite.
- Fábrega et al., VRLog, arXiv:2503.03974 (2025) — verifiable VR; MUST cite.
- Levine & Perez, EPB security policy piece (2021) — notes absence of uniform standards.
- Green, "Critique of Networked Election Systems," MDPI Information 17(1):10 (2025) — claims "first comprehensive synthesis"; MUST read and distinguish (vulnerability survey, not assurance/evidence-gap analysis).
- Stark & Wagner, "Evidence-Based Elections," IEEE S&P 10(5) 2012 — framework anchor.

Novelty delta and confidence

EPB academic security analysis: OPEN (verified 2026-08-29; an APSA preprint itself states "there is little research on the risk associated with e-pollbooks"). Unified-stack assurance-gap analysis: OPEN. VR certification-traceability layer: OPEN (VR security generally is PARTIALLY TAKEN — claim only the traceability layer). Confidence: verified for the gap as scoped; re-verify against USENIX Security 2026 / NDSS 2026 programs before submission.

Threat model

No adversary in the loop for the study itself. Modeled adversary: supply-chain/insider attacker whose capabilities are bounded by what each tier's testing can detect. Trust boundary: certified configuration vs field-deployed reality; transitive trust of machines and EPBs on VR data integrity.

Testability and safety

Entirely public documents and lawful web archives. No contact with any election system, vendor, or government network. No exceptions.

Independent ground truth

Court-filed and CISA-confirmed failure classes; normative requirement text; independent re-coding of a sample; EAC public statements as comparator claims.

Baselines

EAC public adequacy statements; TTBR/EVEREST qualitative gap narratives; Green 2025 synthesis (after differentiation).

Ablations

VVSG 1.0 vs 2.0; with/without state-level evidence; EPB voluntary vs hypothetical mandatory requirements.

Metrics

Per-tier requirement coverage per failure class; traceability-graph coverage; inter-coder ; evidence-opacity rate; VVSG 1.0→2.0 coverage delta.

Negative result

Certification evidence covers the documented failure classes → strong positive assurance finding; equally publishable.

Main risk

Public test reports too redacted/summary-level → mitigate via RfIs, state reports, ESTEP pilot report; residual opacity becomes the evidence-opacity finding.

Kill criterion

If <20% of security requirements have enough public evidence to classify either way, reframe as an evidence-opacity measurement paper or abandon.

Candidate venues (ranked)

1. USENIX Security / EVT workshop (entry point). 2. E-Vote-ID. 3. IEEE Security & Privacy (policy-facing version) + CRA CCC/NIST-style white paper as parallel output.

Phase-2 open items

Final novelty re-check (2026 conference programs; read Green 2025 in full); corpus acquisition plan with lawful archiving; coding schema draft; adjudication protocol; statistics plan (, coverage CIs); artifact plan (public corpus + graph + scripts).